# Siming (Andy) Fu

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## EDUCATION

**Washington University in St. Louis | St. Louis, MO** Aug 2023 - May 2027

B.S. in Electrical Engineering | GPA: 3.80/4.00

Coursework: Signals and Systems, Probability and Statistics, Differential Equations and Dynamical Systems Modeling

## SKILLS

**Languages:** Python, Java, C, SQL, MATLAB, Arduino C

**Software and ML:** Linux/Unix, Git, ROS 2, PyTorch, TensorFlow, OpenCV, YOLO, Gazebo, RViz

**Systems and Engineering:** REST APIs, multithreading, CAN, DBC files, Raspberry Pi, sensor integration, data acquisition

## INTERNSHIP EXPERIENCE

**Robotics Software Engineering Intern**, *Innovation AI | San Jose, CA / Remote* Jun 2026 - Present

- Designed a modular ROS 2 Humble system in Python with separate description, perception, control, and bringup packages, enabling robot models, sensing, tracking, simulation, and test utilities to be developed and validated independently.
- Integrated GPU-enabled YOLO and OpenCV person detection with temporal smoothing, target-state estimation, closed-loop two-axis gimbal tracking, and mobile-base control; added confidence checks, hysteresis, acceleration limits, stale-message timeouts, and target-loss safety stops.
- Built repeatable moving-target simulations and synchronized logs for confidence, target state, velocity commands, and odometry; used Linux, Git, ROS 2 CLI, TF inspection, topic-rate measurements, and controller feedback to debug cross-node integration issues.

**Software Engineering Intern**, *AllPaid Inc. | Indianapolis, IN / Remote* Jun 2025 - Sep 2025

- Built an internal API that detected long-running payment-processing messages, queried supporting SQL records, and escalated qualified cases to the QA team for investigation.
- Translated QA workflow requirements into database validation rules, escalation conditions, and edge-case handling, linking runtime messages to related transaction records with clear and repeatable decision logic.
- Documented expected inputs and outputs, database checks, failure scenarios, and handoff behavior so the workflow could be reviewed, maintained, and extended within an existing payment-software environment.

**Electrical Engineer**, *Glorysky Electronics Co., Ltd. | Huizhou, China* Jul 2025 - Sep 2025

- Collaborated with senior engineers on schematic reviews, circuit troubleshooting, component-level checks, and documentation of test observations for consumer-electronics projects.
- Contributed to PCB design and review tasks, connecting electrical requirements and schematic intent to layout decisions, manufacturability constraints, and assembled-hardware behavior.
- Supported the workflow from customer requirements through CAM/MI preparation, fabrication, assembly, testing, and manufacturing handoff; organized engineering notes and test results for cross-functional review.

## PROJECT EXPERIENCE

**Research Assistant**, *AXIS Lab, Washington University in St. Louis* Sep 2025 - Present

- Reproduced the published EnerCage-HC2 wireless-power-transfer architecture in Ansys HFSS as a reference model, then defined a 100 mm-diameter cylindrical homecage system at 13.56 MHz with a receiver coil of 10 mm or less for mouse-mounted optogenetic implants.
- Designed and simulated six cylindrical resonator variants; showed that tilted rings and receiver-side resonators did not improve coupling, guiding a dual-horizontal-resonator design with peak |S21| of −17.6 dB and 1.1 dB lateral / 0.7 dB vertical variation (~16× better vertical uniformity than a single ring).
- Estimated ~3 W of ideal transmitter input to deliver a 50 mW load at the simulated coupling level; documented assumptions, negative results, and matching-network, rectifier, and validation limitations to guide coil fabrication and experimental testing.

**3D Attention U-Net for Volumetric CT Segmentation**, *Computational Imaging Group* Jan 2025 - Jun 2025

- Designed and implemented a PyTorch 3D Attention U-Net with volumetric skip connections and attention gates, replacing a 2D TensorFlow/TFRecord workflow with direct HDF5 volume loading and 5D tensor processing.
- Built preprocessing, combined cross-entropy/Dice loss, voxel-wise metrics, prediction outputs, and a standalone checker; instrumented pipeline stages to remove a one-hot-encoding bottleneck and used cropped volumes and mixed precision to fit training within GPU memory limits.

**Electronics and Data Acquisition**, *WashU Racing Club* Sep 2023 - Present

- Developed Arduino C firmware for control and sensor boards and integrated CAN messages and DBC definitions to map message IDs, signal scaling, units, and telemetry across vehicle electronics.
- Configured AiM Dashboard and Race Studio data workflows, analyzed test-lap logs, and developed bench, integration, and full-vehicle validation protocols for power-distribution, sensor, and control-board testing.